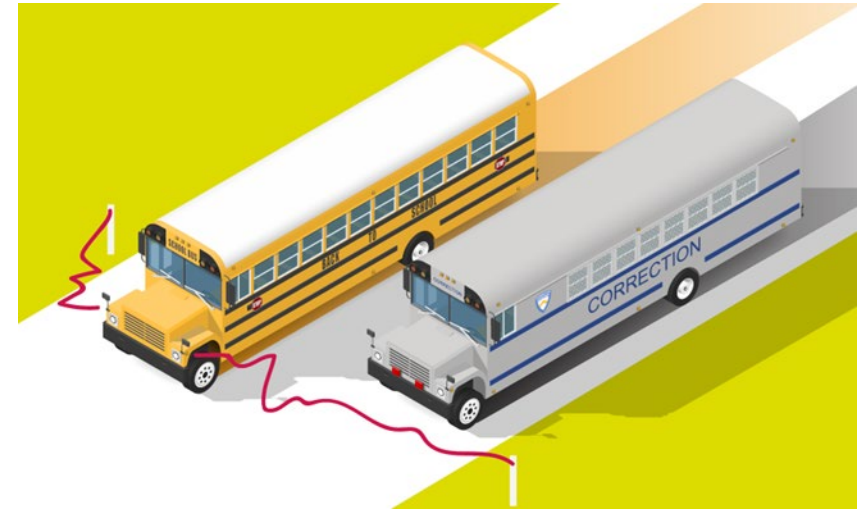


Use of IV with hierarchical analysis/ sample frame segmentation to examine crime reducing policy alternatives to reduce crime among adolescents



Geek Seminar: Queensland University of Technology
February 4, 2026

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⁵ [World Wellbeing Panel](#)

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BITA

ARC Training Centre for Behavioural
Insights for Technology Adoption



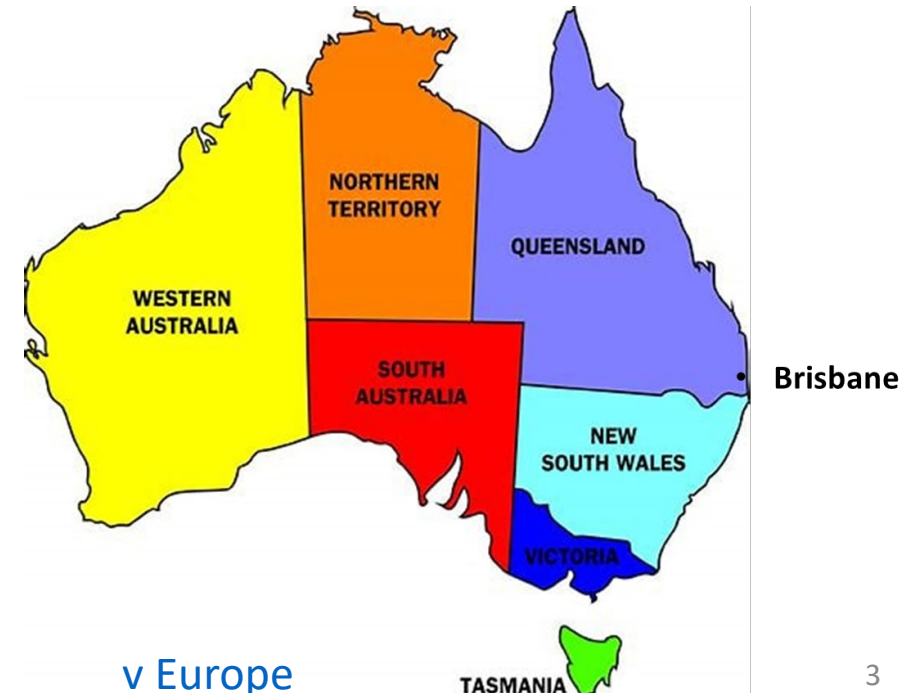
BEST CENTRE
Behavioural Economics . Society . Technology

Is an extra year at the beginning or end of schooling better for reducing youth crime?

Study Data

- Queensland Department of Education Data:** Population school administration 2015-2020 data (176,929 individuals; 83,235 females, aged 15 to 18 years) linked by postcode to:
- Australian Curriculum Assessment Reporting Authority (ACARA) school data
 - Index of Community Socio-educational Advantage (ICSEA): Index of Relative Social Disadvantage (IRSD) (ABS, 2016);

Crime Data: Queensland Police Service Crime 2015-2020 data linked to the education data at individual level.



Sample Frame:

1. **Extra Year of Schooling (2006) at the end of School:** Year 2002 to 2013; 15 to 21 year old, male and female Queensland State School Students.
2. **Extra Year of ECEC Schooling (2006) at the beginning of School:** Year 2015 to 2020; 15 to 18 year old, male and female Queensland State School Students.

Method & Models

Treatment: Universal introduction of a ‘High Quality’ Preparatory Year of Education (*SL*) in the Australian State of Queensland, 2007; uptake “104%” (DoE, 2009).

Analytical Methods: Sharp Regression Discontinuity Design (Imbens and Lemieux, 2008); STATA: ivreg2 (Baum, Schaffer & Stillman, 2007, 2022):

$$O_{it} = \beta_1 S_{it} + \gamma X_{it} + f(a, t) + \varepsilon_{it} \quad \text{Linear model; endogeneity}$$

$$O_{it} = \theta_1 SL_{it} + \pi_1 X_{it} + \beta D_{ijt} + f(r, y) + \varepsilon_{it} \quad (1) \text{ OLS}$$

$$S_{it} = \theta_2 SL_{it} + \pi_2 X_{it} + \beta D_{ijt} + f(r, y) + v_{it} \quad (2) \text{ stage 1}$$

$$O_{it} = \delta_0 + \delta_1 S_{it} + \delta_2 X_{it} + \beta D_{ijt} + f(r, y) + v_{it} \quad (3) \text{ stage 2}$$

Life Domains (D_{ijt}) where our children spend their time: Community, Family, School, Classroom.

Causal estimates of education are derived from the structural form (4): IV estimate of the impact of schooling ($\delta_1 S_{it}$) on crime outcomes (O_{it}) is given by the ratio of the reduced form coefficients in (2) and (3), θ_1/θ_2 .

Results (males): Beaton, Kidd, Machin & Sarker, 2018

Table 8

Change in individual offender behaviour before and after reform.

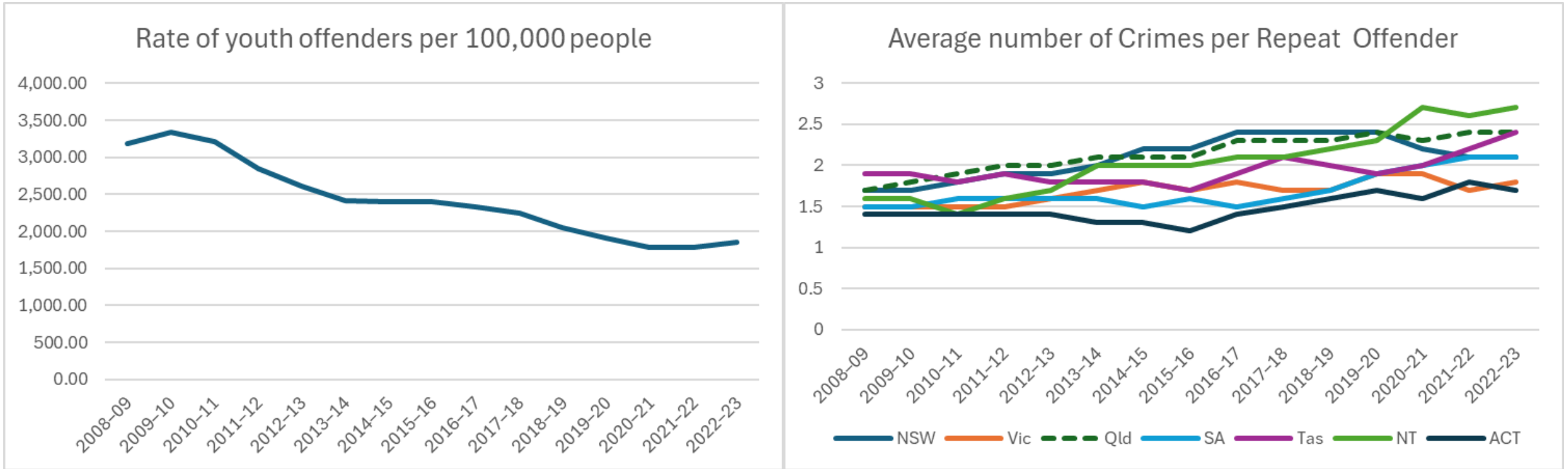
	Discontinuity ± 3 , ages 15–21, cohorts 1987 to 1992, years 2002 to 2013								
	Ever Offends X 100			Offends More Than One Year X 100			Offends Five or More Years X 100		
	Violent Crime	Property Crime	Drug Crime	Violent Crime	Property Crime	Drug Crime	Violent Crime	Property Crime	Drug Crime
A. Males									
Earning or Learning Reform	–0.546 (0.119)	–2.480 (0.168)	–2.411 (0.146)	–0.218 (0.871)	0.673 (0.652)	–1.994 (0.744)	–0.000 (0.166)	–0.051 (0.284)	0.033 (0.137)
Number of Individuals	151,709	151,709	151,709	9305	20,124	14,334	9305	20,124	14,334
Pre-Reform Mean	6.429	14.616	10.764	26.225	36.622	32.393	0.723	4.633	7.150
B. Females									
Earning or Learning Reform	–0.083 (0.070)	–0.778 (0.123)	–0.745 (0.083)	1.779 (1.357)	1.009 (0.813)	0.100 (0.125)	0.076 (0.182)	–0.241 (0.242)	0.106 (0.190)
Number of Individuals	147,079	147,079	147,079	2978	9334	3997	2978	9334	3997
Pre-Reform Mean	2.069	6.777	3.124	17.742	20.834	21.115	0.228	1.676	3.711

Notes: Standard errors in parentheses.

Policy Summary: Extra Education reduces crime, but not for the persistent offenders.

- So, how do we improve the situation?
 - Consider policy alternatives
 - Find evidence to support them in a **target** sub-population.
 - e.g persistent offenders.

Youth offender rates rise for first time in more than a decade, with recidivism rates also up, data shows (ABC News, 8 May 2024; ABS)



OVIEDO, Spain (Alyssa McMurtry, 7.09.2023) [Spain reports ‘alarming’ surge in youth crime \(aa.com.tr\)](https://www.aa.com.tr/en/News/Spain-reports-alarming-surge-in-youth-crime/2023-09-07)

- Spain’s State Attorney General's Office warned of an **“alarming” increase in crimes committed by youth** in an annual report released on Thursday.

[Germany sees rise in crime among children – DW – 03/30/2023](https://www.dw.com/en/germany-sees-rise-in-crime-among-children/a-63444444)

- The number of **crimes committed by children in Germany has risen by a third in 2022** compared to the previous year, according to statistics released by the Federal Criminal Police Office (BKA) on Thursday.

Should we be satisfied with the current situation?



Crime	01/12/20	01/12/21	% Change
Assault	1849	2653	+43.48
Domestic Violence	1139	1751	+53.73
Alcohol Related	903	1396	+54.6
Sexual Assault	88	79	-10.23

[Northern Territory Government, Australia](#)

Queensland to introduce more severe criminal penalties for young offenders

2021: 162,059 persons in Toowoomba (ABS,2021)

22,688 persons aged 10 to 20

1862/22688= 8.2%

Policy Summary: Extra Education reduces crime, but not for the persistent offenders.

- So, how do we improve the situation?
 - Consider policy alternative
 - Find evidence to support them in your target population.

Policy Alternative/ Consider the Literature: **extra year at the beginning of schooling?**

- **Focussed Early Childhood Education & Care (ECEC)** programs (Chicago Child–Parent Centers, High/Scope Perry Preschool Program, and the Carolina Abecedarian Project) provide significant evidence; **lifetime returns are highest when education interventions occur in early childhood** (Anderson, 2008; Gramlich, 1986; Heckman & Karapakula, 2019; Temple & Reynolds, 2007).
- **Universal ECEC access programs are less effective than programs targeted to individual child needs; positive effects fade over time** (Warren, O'Connor, Smart & Edwards, 2016).
- **Dosage matters**: full-day preschool participants had higher test scores for socioemotional development, language, math & physical health, but not literacy and cognitive development; probably explainable by higher attendance and **parental involvement**, which led to fewer chronic absences (Reynolds et. al., 2022), and; less likely to be abused or commit crimes (Arteaga et. al., 2014).
- **High Quality preschool is beneficial to young children** (Morgan, 2019) and appears to **predict later academic performance** and adult outcomes (Tayler et. al., 2013; Warren et. al., 2016).
- **High Quality ECEC reduces crime**; more for girls; more for **most disadvantaged** (Garcia, Heckman & Ziff, 2019).

So, Higher quality ECEC improves academic performance?

- Does it do it for our target population?

Sample Frame:

3. Year 2004 to 2012; 5 to 18 year old, male and female Queensland State School Students.

Treatment: ECEC since 2008

School Outcomes: Prep improves school attendance, effort, behaviour & achievement in grade 3, 6 & 10

(Thorpe, Potia, Beaton, Rankin & Staton, 2020)

Male Students

Year Level	3	6	10
Attendance Rate			**
Behaviour			
English	**	**	**
Science	**	**	**
Mathematics	**	**	
Effort			
English	**	**	**
Science	**	**	
Mathematics	**	**	**
Achievement			
English	**	**	
Science	*	**	**
Mathematics	**	**	**

Male Disadvantaged Students

Year	3	6	10
Attendance			
Attendance Rate	+	**	
Behaviour			
English	**	+	+
Science	**	+	
Mathematics	**	+	
Effort			
English	*	*	
Science	*		
Mathematics	+	**	
Achievement			
English		**	
Science		**	
Mathematics		**	

Most Disadvantaged

Summary of RDD structural form (4) results for year 3, 6 & 10 male students: positive coefficient (green), negative coefficient (red); non-significant (blank), + 10%, * 5%, and ** 1% or lower.

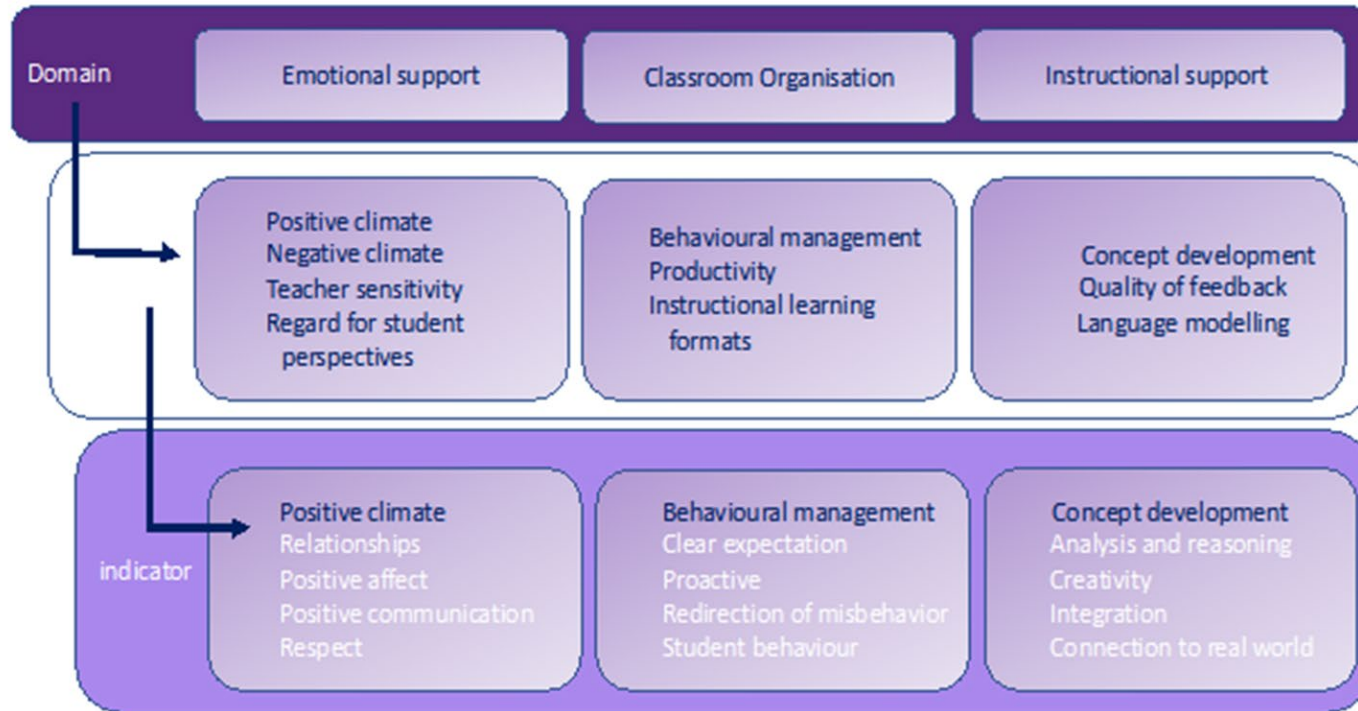
Females

So, what, it didn't help the disadvantaged kids

- Why?

ECEC quality contributes to education outcomes:

Effective Early Learning Experiences for Children e4Kids (Tayler, Ishimine, Cloney, Cleveland & Thorpe, 2013)



Pianta, La Paro, and Hamre, 2008, figure 1.7:
ECEC quality outcomes based on CLASS
observations – Domain and Dimension

Educational Outcomes of Queensland's Investment in Early Childhood Education and Care 2007-2020 (Thorpe, Potia, Beaton, Rankin & Staton, 2020).

e4Kids & DoE data linked n=4,636; 692 individuals;

assessed the quality of ECEC provision using intensive observation to identify key elements of ECEC programs that predict educational outcomes.

Sample Frame:

4. **2010 e4Kids Data Source** contains 692 children who can be linked to the 2015 to 2020 DoE data, sample size of $n=4,636$.

ECEC quality contributes to education outcomes:

Educational Outcomes of Queensland's Investment in Early Childhood Education and Care 2007- 2020 (Thorpe, Potia, Beaton, Rankin & Staton, 2020).

e4Kids & DoE data linked;
n=4,636; 692 individuals;

CLASS Dimensions;
assessed the quality of ECEC provision using
intensive observation to identify key elements of
ECEC programs that predict educational
outcomes.

		<u>Key Learning Area Outcomes*</u>		
		English	Science	Mathematics
CLASS		.0512 ⁺	.0492 ⁺	.0567 ⁺
	Classroom Organization	-0.0591	0.0223	-0.0552
	Behavior Management			
	Productivity	.1121 [*]	.1308 ^{**}	0.0532
		-.1016 [*]	-.0808 ⁺	-0.0461
	Instructional Learning Formats	-0.0182	0.0113	-0.0401
	Emotional Support	.1202 [*]	.0795 ⁺	.1767 ^{***}
	Positive Climate	-.0776 ⁺	-.0751 ⁺	-.1061 [*]
	Reverse Negative Climate	0.025	-0.0291	0.075
	Teacher Sensitivity	-0.0062	0.0097	0.0436
	Regard for Student Perspectives	.0923 ^{***}	.0756 ^{**}	.1102 ^{***}
	Instructional Support	0.0043	-0.0189	-0.0496
	Concept Development	-.1259 ^{**}	-.1647 ^{***}	-.2093 ^{***}
	Quality of Feedback	.1769 ^{***}	.1612 ^{***}	.2122 ^{***}
	Language Modeling	-0.0436	-0.023	-0.0407

* Controls: gender, socio-economic and age and year level fixed effects

The sybols +, *, **, *** represent statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively

Policy Summary: Early Childhood Education & Care (ECEC) is important?

Quality of ECEC, as assessed through observation of classroom interactions (e4Kids: Tayler, Ishimine, Cloney, Cleveland & Thorpe, 2013) were found to be significantly and enduringly associated with effort, behaviour and educational attainments even when controlling potential selection effects of family background (Thorpe, Potia, Beaton, Rankin & Staton, 2020).

- School engagement as assessed by teacher rated effort and behaviour, are strong predictors of educational attainment.
- Better regulated behaviour and greater effort are associated with ECEC interactional quality (**Improved Behavioural Management**) at age 3-4 years; specifically, the CLASS:
- Train ECEC Educators to Improve: **Behaviour Management & Regard for Student Perspectives**

Is an extra year at the beginning or end of schooling better for reducing youth crime?

If High Quality ECEC improves school attendance, effort, behaviour & achievement in grade 3, 6, what happened to Youth Crime?

Hypothesis: Additional year of High Quality Preparatory Education (Prep) prior to grade 1 reduces 15-to-17-year-old youth crime.

But will ECEC reduce Youth Crime?

Sample Frame:

- 3. Extra Year of ECEC Schooling (2006) at the beginning of School:** Year 2015 to 2020; 15 to 18 year old, male and female Queensland State School Students.

Results: Youth Crime

Why?

	<u>P(Committing an Offence)</u>			<u>Number of Offences</u>		
	(4)	(5)	(6)	(7)	(8)	(9)
	All	Male	Female	All	Male	Female
Stage 1 DV: Years of Education						
Preparatory Year of Education Treatment	0.841**	0.838**	0.843**	0.841**	0.838**	0.843**
	(0.002)	(0.004)	(0.004)	(0.002)	(0.004)	(0.004)
F-stat. ³	1.2e+05	59156.29	57648.72	1.2e+05	59156.29	57648.72
Stage 2						
Years of Education	-0.0715**	-0.0983**	-0.0345**	-0.1114**	-0.1557**	-0.0425**
	(0.0017)	(0.0027)	(0.002)	(0.0101)	(0.0171)	(0.0087)
Fixed Effects						
Year	yes	yes	yes	yes	yes	yes
Relative age	yes	yes	yes	yes	yes	yes
Number of Individuals	176,929	93694	83,235	176,929	93694	83,235
N	489466	255373	234,093	489466	255373	234,093
Under identification test ¹	1.2e+04	2.5e+04	2.2e+04	1.2e+04	2.5e+04	2.2e+04
Weak identification test ²	4.7e+05	5.90e+04	5.8e+04	4.7e+05	5.9e+04	5.8e+04
Marginal effects; Standard errors in parentheses; + p<0.10, * p<0.05, ** p<0.01						
¹ Kleibergen-Paap rk LM statistic; ² Kleibergen-Paap rk Wald F statistic						
³ Exceeds F-stat requirements per Lee et. al., 2021						

End of School (Beaton et al., 2018):

-0.062**

-0.099**

-0.0219**

-0.06** 23

Policy Alternative/ Age Literature; Red Shirt ing

The reasons parents delay sending children to school (ABC Online News; 27 May 2025)

"For me, [the motivation to delay] was definitely more the emotional side of things."

"I also worried about the social aspects, and also fatigue"

"If she was starting school when [she was] four and a half, she would just be too burnt-out."

"A lot of international research says academic competitiveness is why you might hold a child back, but there wasn't a sense [parents in Australia] were making those choices for **competitive reasons**," Dr Van Bergen, Associate Professor of Psychology in Education, Macquarie University.

- **School starting age matters** (Beatton, Kidd, Niu & Vella, 2023); later school entry improves social behaviour, crime reduction (Anderson, 2014; Landersø, Nielsen & Simonsen, 2017). Start school later? (Black, Devereux, Salvanes, 2011).
- **'Opaqueness' of positive results from universal ECEC programs:**
 - positive ECEC effects on grades came from an **(absolute) difference in school starting age** (Drange, Havnes, & Sandsør, 2016);
 - **Month (absolute) of birth** (Crawford, Dearden & Greaves, 2014);
 - Beatton, Kidd, Niu & Vella (2023); **Relative Age difference in enrolment cohort.**

Sample Frame: Beatt on, Kidd & Sanding 2024

- 3. Extra Year of ECEC Schooling (2006) at the beginning of School:** Year 2003 to 2015; 15 to 18 year old, male and female Queensland State School Students.

Hypotheses

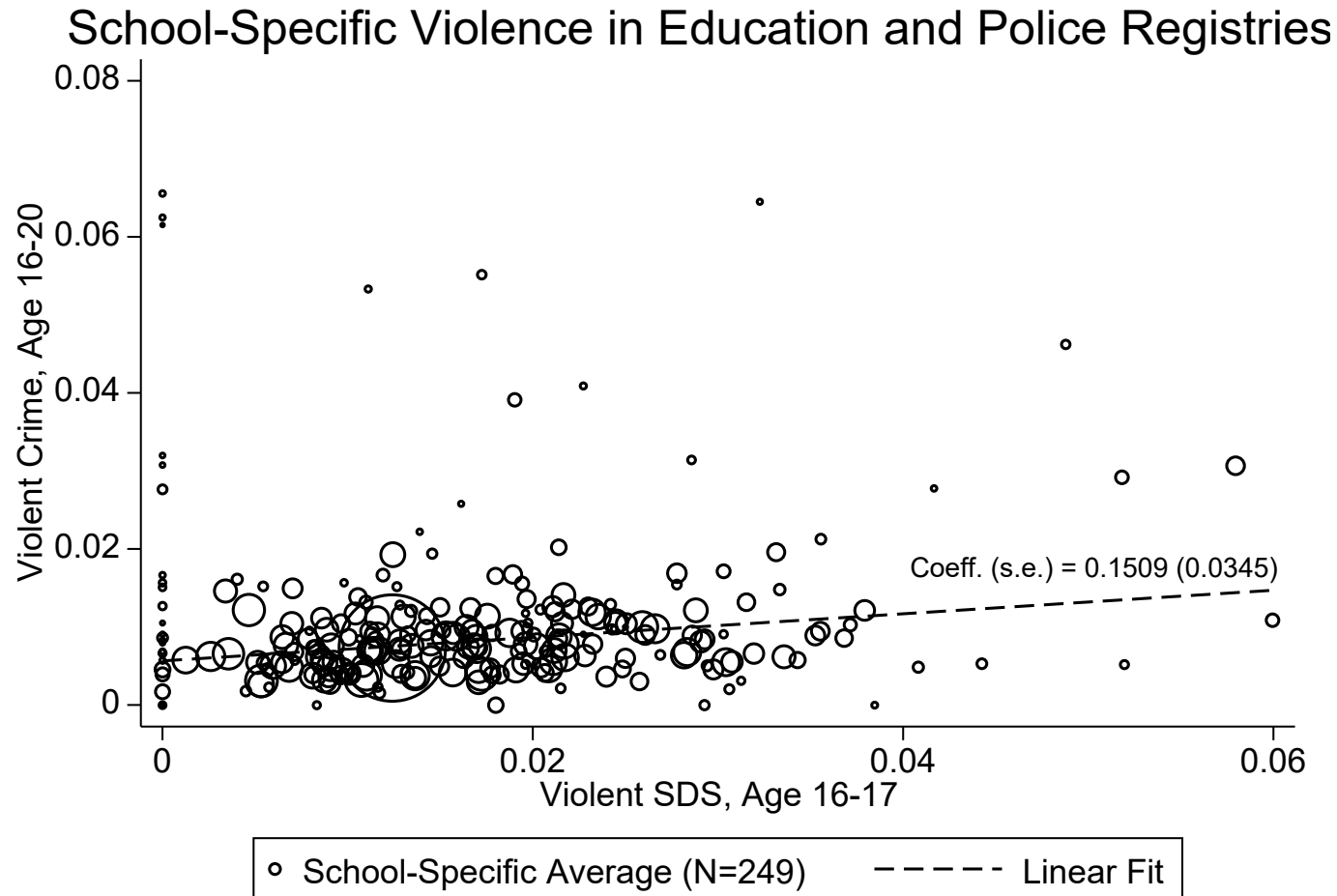
1. Younger kids in the classroom get disciplined more often
2. Kids who get disciplined at school are more likely to commit crime.

Method Change:

- Fuzzy RDD

$$Y_{ict} = \alpha + \beta_1 * ELic + \beta_2 * Yobic + \beta_3 * Yobic * ELic + f(a, t) + \delta * X_{ict} + \vartheta_s + \omega_{ict}, (1)$$

Update or Analytical model to include relative age.



Younger kids get disciplined more often

Table 2. School Disciplinary Sanctions (SDS) and Crime Offences in 1987-89 Birth Cohorts

	(1)	(2)	(3)	(4)	(5)	(6)
	Age 16 to 20	Age 16	Age 17	Age 18	Age 19	Age 20
A. Violent SDS Received						
Number	3,561	2,374	1,187			
Share of Age 16 to 17	1	0.6667	0.3333			
% Rate	1.59%	2.12%	1.06%			
B. Any SDS Received						
Number	4,927	3,276	1,651			
Share of Age 16 to 17	1	0.6649	0.3351			
% Rate	2.20%	2.92%	1.47%			
Juveniles		112,083				
Observations		224,036				

Policy Summary: Extra Education reduces crime, but not for the youngest in the classroom offenders.

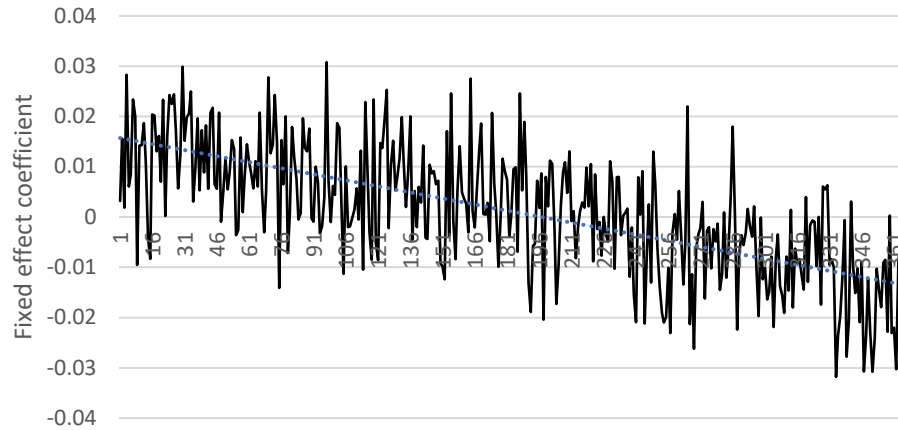
- So, how do we improve the situation?
 - Provide teachers with extra assistance in a classroom with high numbers of younger students .
 - Teachers aid handles behavioural management issues which frees up the teacher to teach.
 - This also helps the remainder of the class improve their learning.
- **Methodology**: Consider relative age when we do crime analysis

How does this change how we consider crime analysis?

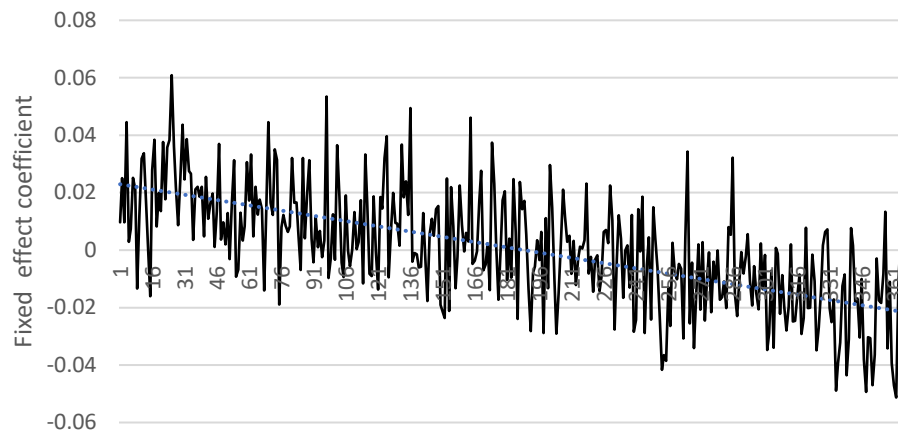
Prep: extra year at the beginning of schooling

Relative age matters

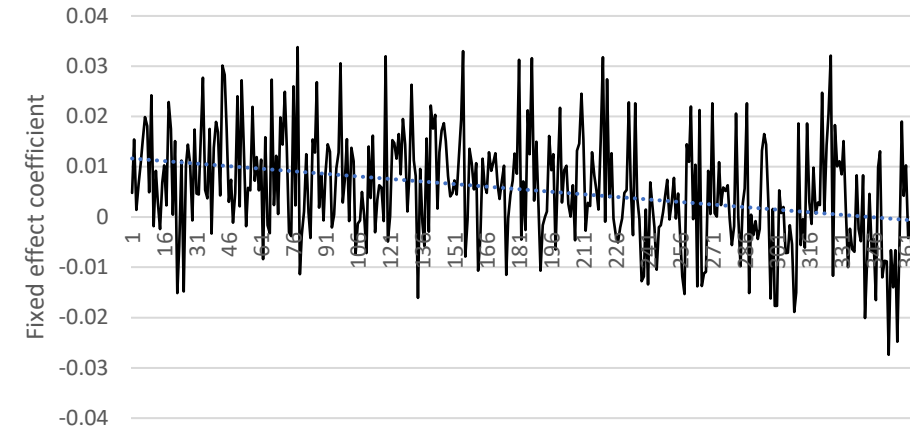
Birthday in Year Fixed Effects on Probability of Committing a Crime - Females & Males



Birthday in Year Fixed Effects on Probability of Committing a Crime - Males



Birthday in Year Fixed Effects on Probability of Committing a Crime - Females



(Beatton, Kidd & Sandi, 2023;
Beatton, Kidd, Niu & Vella, 2023;
Beatton, Potia, Rankin, Staton & Thorpe, 2022;
Thorpe, Potia, Beatton, Rankin & Staton, 2020;
Landersø, Nielsen & Simonsen, 2017;
Drange, Havnes, & Sandsør, 2016;
Crawford, Dearden & Greaves, 2014;
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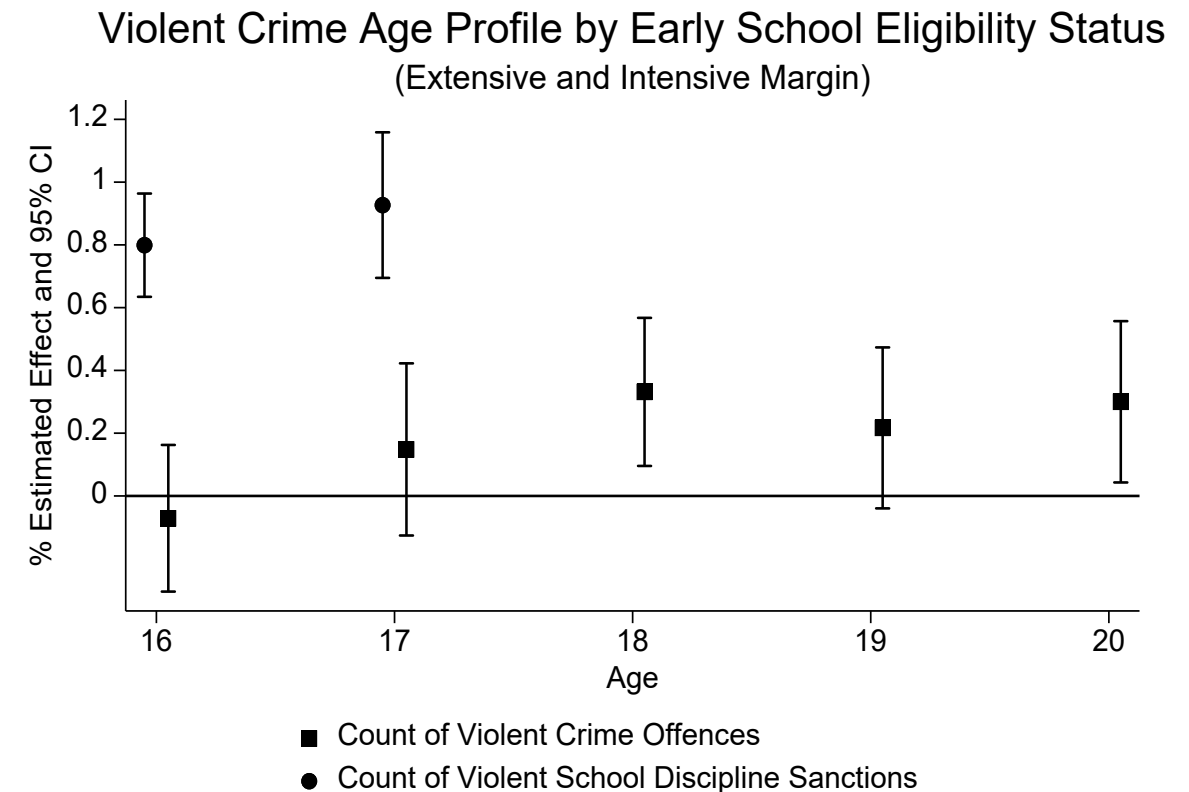
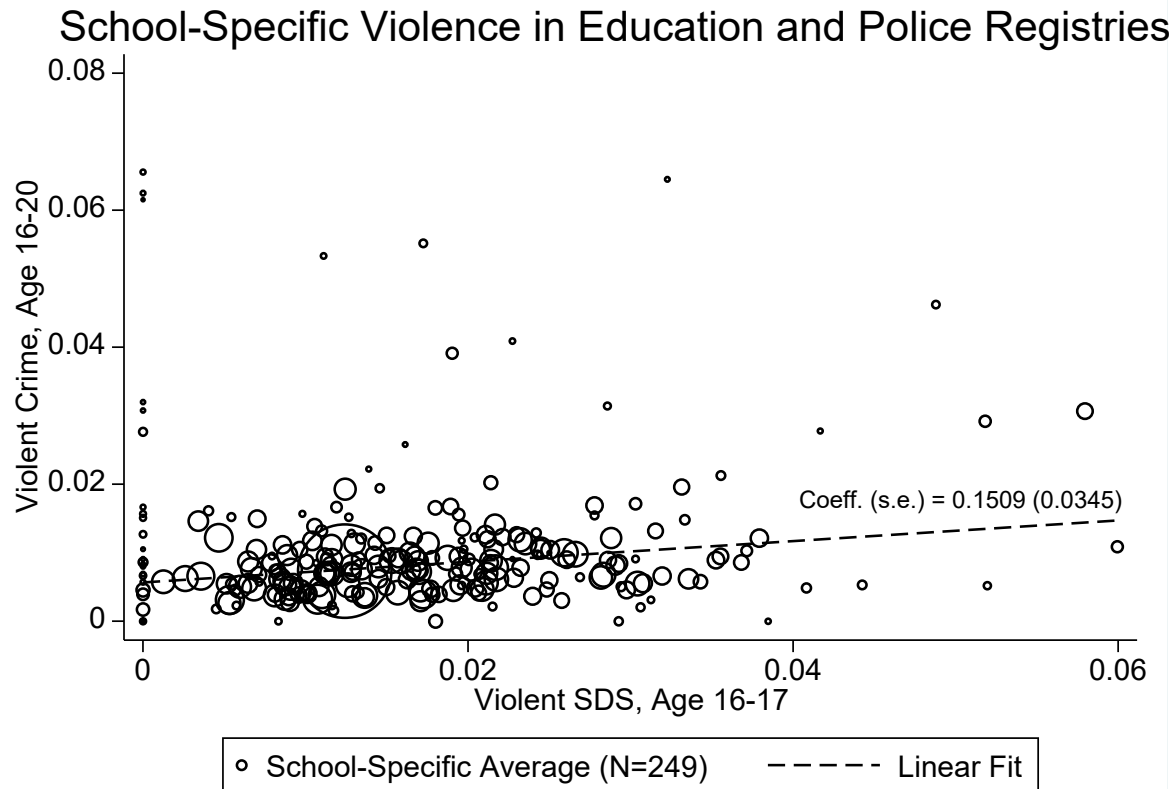
Are there other things that could have an effect on kids committing crime?

Literature

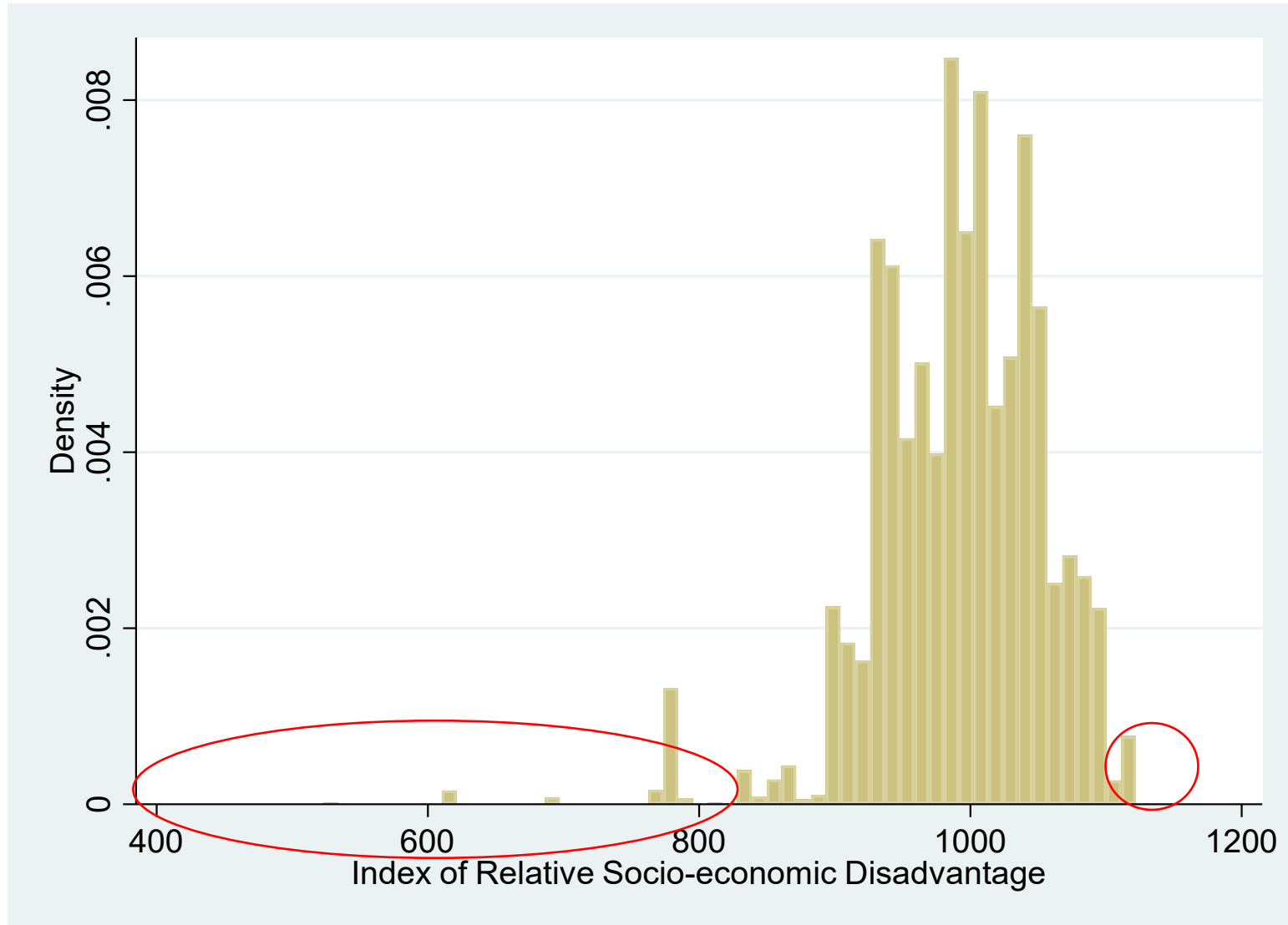
- **Indicators of family disadvantage** tend to have significantly **greater effects on school outcomes** for **boys** (Autor, Figlio, Karbownik, Roth & Wasserman, 2019).
- **School Attendance** strongly affects educational outcomes (Thorpe, Potia, Beatton, Rankin & Staton, 2020), **which affects crime?**
- **Individual Behaviour and Effort** in the **classroom** strongly affects educational **test score outcomes** (Thorpe, Potia, Beatton, Rankin & Staton, 2020 ;Reynolds et. al., 2010), **which affects crime?**

Literature

- **Peer behaviour affects educational outcomes** which affect youth crime (Rotger & Galster, 2019); **in the classroom, school yard & those domains of the wider community where students spend their time?**
- **School disciplinary sanctions in school** predict criminality; older children in an enrolment cohort were less likely to receive disciplinary sanctions at school (Beatton, Kidd & Sandi, 2022). **If we reduce SDS, do we reduce youth crime?**



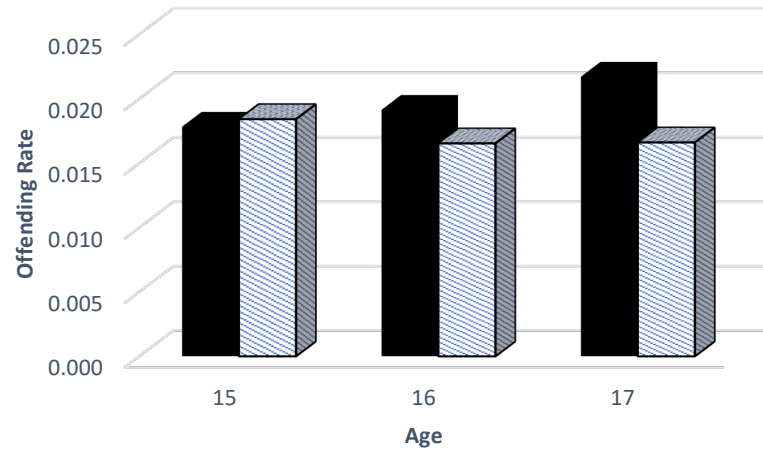
Most Disadvantaged



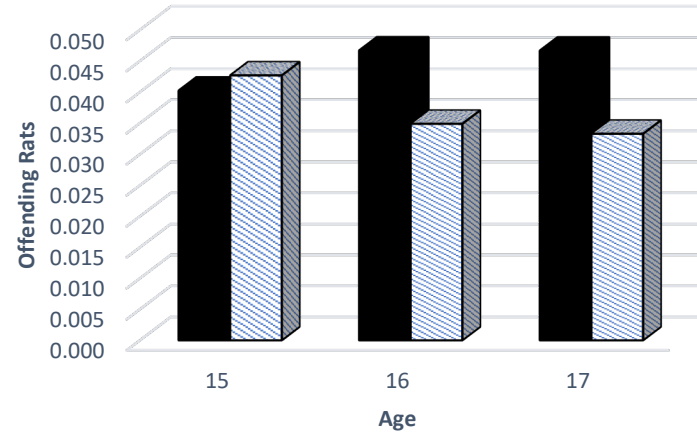
(ABS, 2016)

What happened to Crime post Prep?

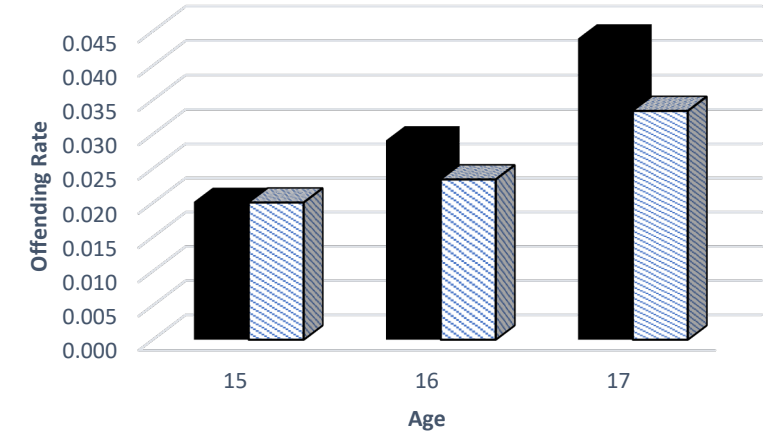
Violent Crime - Males



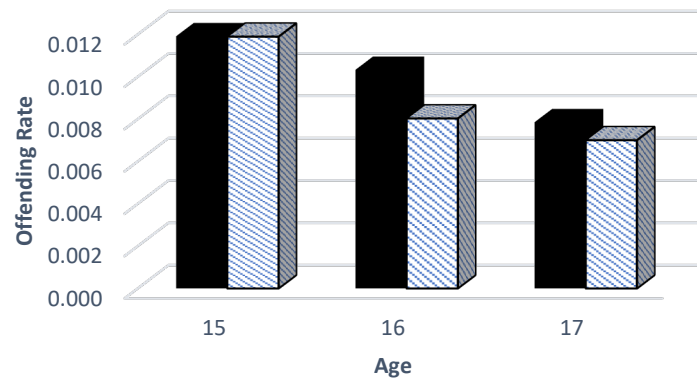
Property Crime - Males



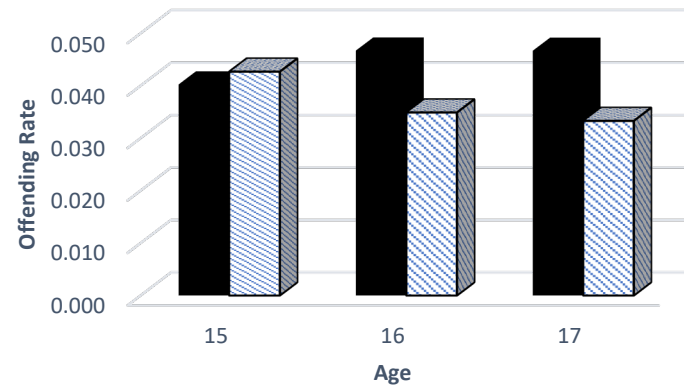
Drug Crime - Males



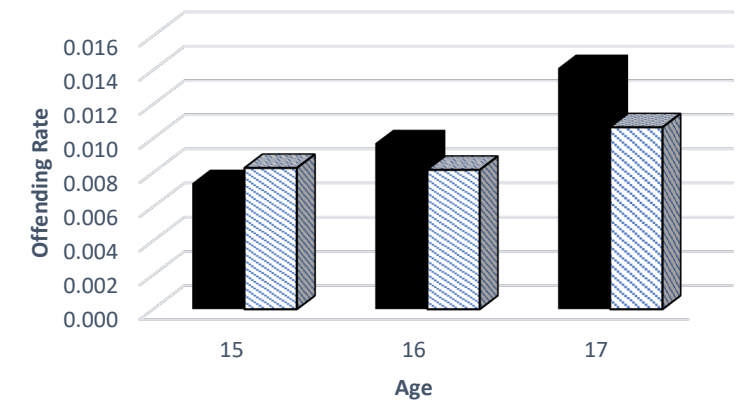
Violent Crime - Females



Property Crime - Females



Drug Crime - Females



End of School

Youth Crime

Why?

	<u>P(Committing an Offence)</u>			<u>Number of Offences</u>		
	(4)	(5)	(6)	(7)	(8)	(9)
	All	Male	Female	All	Male	Female
Stage 1 DV: Years of Education						
Preparatory Year of Education Treatment	0.841**	0.838**	0.843**	0.841**	0.838**	0.843**
	(0.002)	(0.004)	(0.004)	(0.002)	(0.004)	(0.004)
F-stat. ³	1.2e+05	59156.29	57648.72	1.2e+05	59156.29	57648.72
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Weak identification test ²	4.7e+05	5.90e+04	5.8e+04	4.7e+05	5.9e+04	5.8e+04
Marginal effects; Standard errors in parentheses; + p<0.10, * p<0.05, ** p<0.01						
¹ Kleibergen-Paap rk LM statistic; ² Kleibergen-Paap rk Wald F statistic						
³ Exceeds F-stat requirements per Lee et. al., 2021						

End of School (Beaton et al., 2018):

-0.062**

-0.099**

-0.0219**

-0.06** 38

Youth Crime: all; most disadvantaged; persistent offenders

	<u>P(Committing an Offence)</u>			<u>Number of Offences</u>		
	(4)	(5)	(6)	(7)	(8)	(9)
-	All	Male	Female	All	Male	Female
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	(0.0017)	(0.0027)	(0.002)	(0.0101)	(0.0171)	(0.0087)

Persistent Offenders (n=162,886)

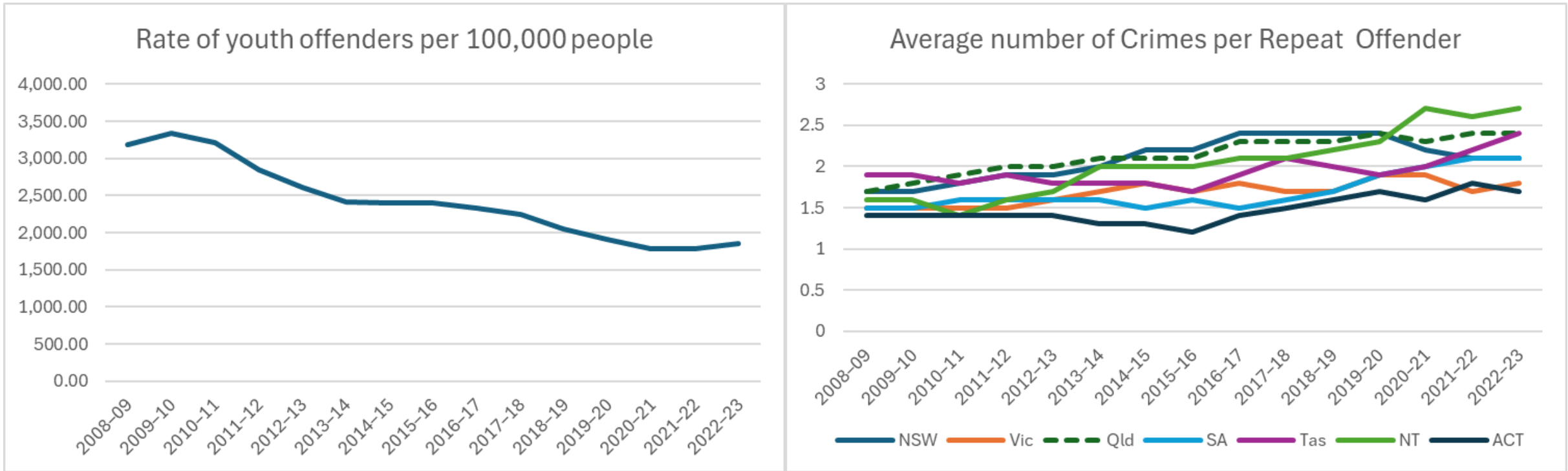
<u>Stage 2</u>						
Years of Education	-0.0370**	-0.0533**	-0.0171**	-0.0544**	-0.1557**	-0.0137+
	(0.0011)	(0.0018)	(0.0011)	(0.0102)	(0.0171)	(0.0081)

Most Disadvantaged (n=4322)

<u>Stage 2</u>						
Years of Education	-0.0902**	-0.0927**	-0.0635**	-0.1121	-0.0478	-0.0050
	(0.0169)	(0.0260)	(0.0198)	(0.1031)	(0.0753)	(0.1781)

Conclusion: An additional year of preparatory education
at the beginning of school reduced youth crime,
including for persistent offenders

Youth offender rates rise for first time in more than a decade, with recidivism rates also up, data shows (ABC News, 8 May 2024; ABS)



OVIEDO, Spain (Alyssa McMurtry, 7.09.2023) [Spain reports ‘alarming’ surge in youth crime \(aa.com.tr\)](#)

- Spain’s State Attorney General's Office warned of an **“alarming” increase in crimes committed by youth** in an annual report released on Thursday.

[Germany sees rise in crime among children – DW – 03/30/2023](#)

- The number of **crimes committed by children in Germany has risen by a third in 2022** compared to the previous year, according to statistics released by the Federal Criminal Police Office (BKA) on Thursday.

Should we be satisfied with the current situation?



Crime	01/12/20	01/12/21	% Change
Assault	1849	2653	+43.48
Domestic Violence	1139	1751	+53.73
Alcohol Related	903	1396	+54.6
Sexual Assault	88	79	-10.23

[Northern Territory Government, Australia](#)

Queensland to introduce more severe criminal penalties for young offenders

2021: 162,059 persons in Toowoomba (ABS,2021)

22,688 persons aged 10 to 20

1862/22688= 8.2%

Should we be satisfied with the current situation?

[Queensland tops nation for child detention and youth repeat offenders, Productivity Commission data reveals \(ABC News, 24 Jan 2023\)](#)

- "Queenslanders spend **\$162 (€ 100) million every year** on youth detention;
- Queensland (pop. 5.3M) also recorded a daily average number of **287 people in youth detention in 2021-22**, again the highest of all states and territories — NSW was second highest at 190 (pop. 8.2M)

The (then) Deputy Premier Steven Miles said today the state's approach worked for:

"90 per cent of offenders"

- Queensland children spent **100,425 nights behind bars** for the year to the end of June 2022.
- Almost two thirds of those nights, **65,298, were Aboriginal and Torres Strait Islander children** in custody.
- Indigenous children represent **less than 10% of the state's population** of people aged 10 to 17.

What more could we do about it?

1. **Universal Access to High Quality ECEC** with a focus on **Behavioural Management, Emotional Support & Consideration of the Individual Child**
 - **Government(s) Response:**
 - [Federal Government](#) (\$AUD 9.5M; (€5.9M)), Queensland & Victoria have been running trials of **Quality** ECEC/pre-school programs; and;
 - [Federal Government](#) **increased ECEC access funding for poorer families**, and; **universal ECEC access bill** (\$AUD 1.8B; (€1.13B))

What more could we do about it:

> School **attendance, effort, behaviour** > **achievement**; < **Crime**?

1. **Access to Preparatory Year** of Early Childhood Education (Prep) improved academic achievement (Thorpe, Potia, Beatton, Rankin & Staton, 2020)
2. **Need to consider a student's Developmental Age; Relative Age within enrolment cohort matters** (Beatton, Kidd & Sarkar, 2015)?
3. Children from **Most Disadvantaged Communities** benefitted much less from Prep academically (Thorpe, Potia, Beatton, Rankin & Staton, 2020), what can we do?

Answers may emerge from examining the Life Domains (D_{ijt}) where students spend their time:

- **Community;**
- **Home/Family;**
- **School, and;**
- **Classroom.**

What could we do about it?

Community on P(Criminality)

<u>Environment:</u>			<u>Community</u>
	Initial analysis (Table 10)		Socio- economic Disadvantage
<u>IV Stage 2:</u>			
Years of Education	-0.0464** (0.0016)		-0.0296** (0.0030)
Index of Relative Socio- economic Deprivation			-0.0001** (0.0000)

- Increase funding to poorer communities

What could we do about it?

Home/Family on P(Criminality);

- **Help Families increase their Income:**
 - Long run: **Improving parent education**
 - **> Job opportunities**
 - **Housing** (UK: Disney et al., 2023)
 - **Decrease net worth poverty** (Gibson-Davis et al., 2025)
 - Short run: **ECEC funding > female labour market participation** (USA: Gibbs et al., 2025)
 - **Increase transfers?** (USA: Noble et. al., 2025)
 - **Reducing Social Safety Nett increases crime** (Denmark: Jinkins et al., 2025)

Environment:		Household	
	Initial analysis (Table 10)	Parent Employment	Parent post high school education
IV Stage 2:			
Years of Education	-0.0464** (0.0016)	-0.0015** (0.0006)	-0.0089** (0.0014)
Parent 1 Employment			
Senior management		-0.0011+ (0.0006)	
Professional		-0.0014** (0.0005)	
Trade		-0.0015** (0.0005)	
Parent 2 Employment			
Senior management		-0.0011 (0.0007)	
Professional		-0.0019** (0.0004)	
Trade		-0.0016** (0.0004)	
Parent 1 Education			
Trade certificate			-0.0033** (0.0010)
Diploma			-0.0037** (0.0011)
Bachelor degree or above			-0.0039** (0.0010)
Parent 2 Education			
Trade certificate			-0.0037** (0.0010)
Diploma			-0.0059** (0.0009)
Bachelor degree or above			-0.0058** (0.0010)

What could we do about it?

School effects on P(Criminality); > school resources < crime

(Beatton, Kidd & Sandi, 2019)

P(Criminality); School Discipline

~8.5% = SDS;

6% SDS=1;

1.5% SDS=2;

0.47% SDS=3;

0.16% SDS=4;

0.37% SDS>=5 (Beatton Kidd & Sandi, 2022)

<u>Environment:</u>	
	School Discipline
<u>IV Stage 2:</u>	
Years of Education	-0.0279** (0.0030)
Committed school offence=1	-0.0318** (0.0015)
Sum annual school offences	0.0016+ (0.0010)

Support first-time offenders to avert in-school re-offending & crime:

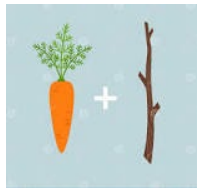
e.g. Queensland Government, Regional Reset: Adopt-a-Cop

What could we do about it?

Classroom effects on P(Criminality)

	(6a)	(6a)	(6a)	(6a)	(6a)
<u>Environment:</u>	School	School	School	School	School
	Classroom Environment	Classroom Environment	Classroom Environment	Classroom Environment	Classroom Environment
IV Stage 2:					
Years of Education	-0.0171** (0.0001)	-0.0002+ (0.0001)	-0.0001+ (0.0001)	-0.0001+ (0.0001)	-0.0001+ (0.0001)
School attendance rate	-0.006** (0.002)				-0.003** (0.0009)
English behaviour		-0.0005** (0.0001)			-0.0002+ (0.0001)
English Effort			-0.0004** (0.0001)		-0.0001 (0.0001)
English Achievement				-0.0002** (0.0001)	-0.0001 (0.0001)

- **Increase school attendance rate** (Population: 87.6%; **Most Disadvantaged Communities 63.5%**);
 - Help improve classroom behaviour;
 - Encourage individual effort;
 - To Improve educational outcomes.



What could we do about it?

A Policy of **Support** not **Punishment**

1. **Positive Behavior Interventions and Supports (PBIS) framework** (Vincent et al. 2015)
2. **Support first-time offenders** to avert in-school re-offending & crime

Funding?

My local primary school gets \$15,451 to educate one child for one school year (2024);

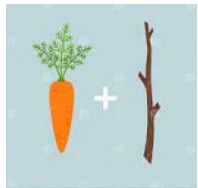
\$75.37 (€ 42.34) per school day

Queenslander budget \$770.9 million (2025-2026) for youth detention;

~\$2162 (€1215) per child/youth per day (average detained/day: 318)

So, what are the Queensland State Government policy responses?

1. The Positive Behaviour for Learning (PBL) framework;
2. **QPS-School programs:** (adopt-a-cop; sport redirection; cultural; activity programs);
3. **2015 Queensland changed how SDS are approved - result: increased SDS** (Beatton, Kidd & Sandi, 2022)
 - **Compulsory sentencing**
 - **Youth detention (no bail recommendations - detaining children in police watchhouses)**
 - **More Police (in schools?)**



Queensland State Government policy responses

Queensland Minister for Youth Justice and Victim Support and Minister for Corrective Services

The Honourable Laura Gerber (Published Tuesday, 24 June, 2025 at 03:17 PM)

Youth Justice, Victim Support and Corrections Budget to help deliver safety where you live

- Budget delivers the foundation for a fresh start with a record investment in **youth crime fighting programs**, support for victims of crime, and strengthened security in our prisons.
- **\$560 million in new early intervention and rehabilitation programs to arrest Labor's Youth Crime Crisis.**
- More support than ever before for victims of crime with a new \$50 million Victims Advocate Service.
- **\$2.387 billion over six years for additional prison capacity.**

Your Policy Suggestions?



BITA

ARC Training Centre for Behavioural
Insights for Technology Adoption



BEST CENTRE

Behavioural Economics . Society . Technology

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Most Disadvantaged v Others

	Others	Most Disadvantaged	t-value	
Individuals	173,820	4322 (2.5%)		
Most Disadvantaged (IRSD)	996	755	437.58	**
Female	0.478	0.481	-0.47	
Number Children in Family	1.539	1.952	-29.94	**
Maximum Education Years	12.298	12.223	10.82	**
<u>Parent 1</u>				
Manager	0.097	0.028	20.53	**
Professional	0.122	0.036	23.44	**
Trade	0.218	0.121	20.61	**
Unemployed	0.183	0.323	-31.76	**
<u>Parent 2</u>				
Manager	0.113	0.029	22.11	**
Professional	0.151	0.056	22.12	**
Trade	0.253	0.136	22.51	**
Unemployed	0.058	0.170	-39.39	**
<u>School</u>				
Attendance	0.876	0.818	40.12	**
Behaviour	4.170	3.751	36.64	**
Effort	3.812	3.417	29.86	**
Achievement	3.416	3.035	33.11	**
P(Disciplined at School)	0.092	0.139	-15.84	**
Average Disciplines/year	0.113	0.180	-14.65	**
+ p<0.10, * p<0.05, ** p<0.01				

ECEC improves school attendance, effort, behaviour & achievement in grade 3.6 & 10 (Thorpe, Potia, Beaton, Rankin & Staton, 2020).

Female Students

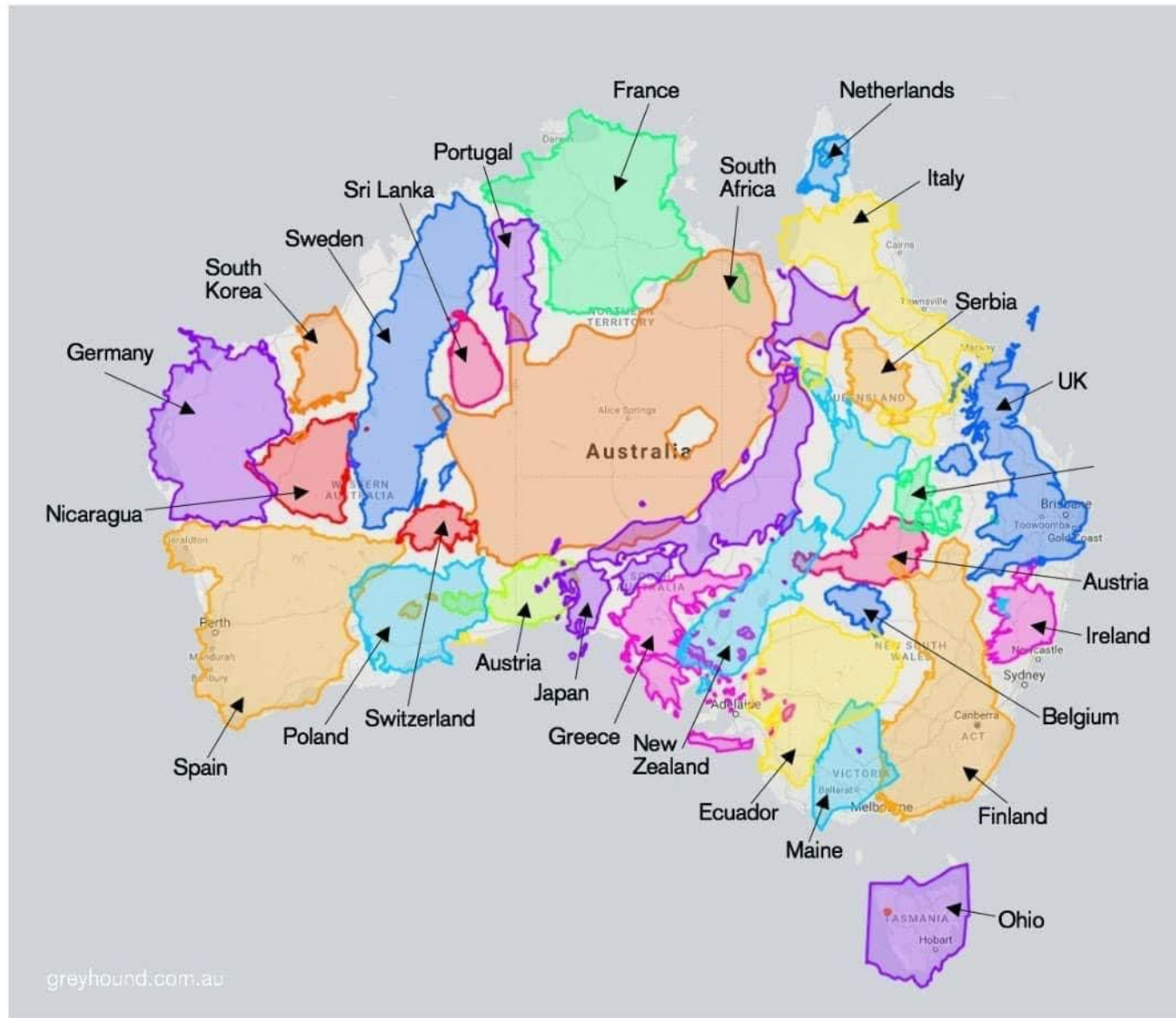
Year Level	3	6	10
Attendance Rate		**	
Behaviour			
English	**	**	**
Science	**	**	**
Mathematics	**	**	
Effort			
English	**	**	**
Science	**	**	**
Mathematics	**	**	**
Achievement			
English	**	**	**
Science		**	**
Mathematics	**	**	**

Female Disadvantaged Students

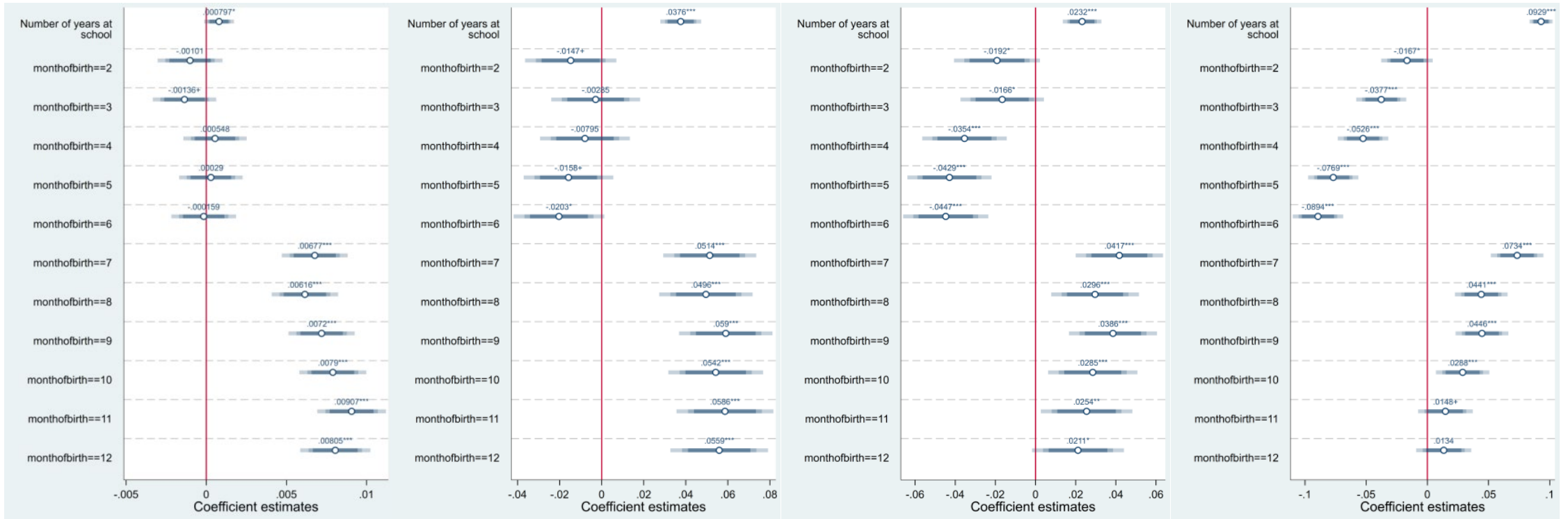
Year	3	6	10
Attendance			
Attendance Rate		**	
Behaviour			
English	+	*	
Science	+	*	*
Mathematics	*	**	
Effort			
English		+	
Science	*		
Mathematics	+	**	
Achievement			
English			
Science			
Mathematics		**	

*Summary of RDD structural form (4) results for year 3, 6 & 10 female students: positive coefficient (green), negative coefficient (red); non-significant (blank), + 10%, * 5%, and ** 1% or lower.*

The true size of Australia 🤯



Relative Age Effects: English Year 6



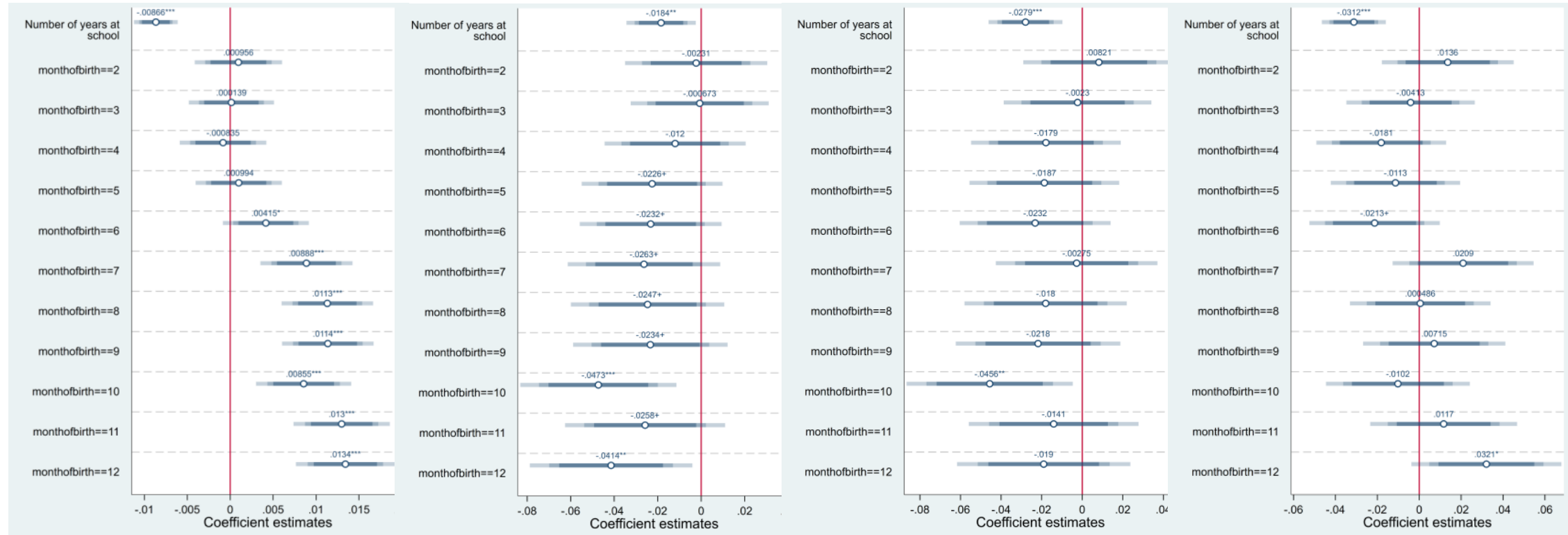
Attendance

Behaviour

Effort

Achievement

Relative Age Effects: English Year 10



Attendance

Behaviour

Effort

Achievement

Peak Youth Crime Discontinuity Sample:

Year	2015	2016	2017	2018	2019	2020
Age	13	14	15	16	17	18
Student Birth Year						
Jan - Dec 2000	10	11	12	12	12	12
Jan - Dec 2001	9	10	11	12	12	12
<i>1 Jan 2002 -30 Jun 2002</i>	<i>9</i>	<i>10</i>	<i>11</i>	<i>12</i>	<i>13</i>	<i>13</i>
<i>Jul 2002 - Jun 2003</i>	<i>8</i>	<i>9</i>	<i>10</i>	<i>11</i>	<i>12</i>	<i>13</i>

Birth Cohort/School Enrolment Group	Age				Total
	15	16	17	18	
2000	40,667	40,354	38,381	11,535	130,937
2001	40,661	40,371	38,101	20,663	139,796
<i>1Jan2002-30Jun2002</i>	<i>20,547</i>	<i>20,159</i>	<i>35,460</i>	<i>1,861</i>	<i>78,026</i>
<i>1Jul2002-30Jun2003</i>	<i>40,734</i>	<i>59,165</i>	<i>39,222</i>	<i>1,586</i>	<i>140,707</i>
Total	142,609	160,049	151,164	35,645	489,466

Birth Cohort/School Enrolment Group	Year						Total
	2015	2016	2017	2018	2019	2020	
2000	40,667	40,354	38,381	11,535			130,937
2001		40,661	40,371	38,101	12,161	8,502	139,796
<i>1Jan2002-30Jun2002</i>			<i>20,547</i>	<i>20,159</i>	<i>18,476</i>	<i>18,845</i>	<i>78,026</i>
<i>1Jul2002-30Jun2003</i>			<i>20,161</i>	<i>41,071</i>	<i>39,495</i>	<i>39,980</i>	<i>140,707</i>
Total	40,667	81,015	119,460	110,866	70,132	67,327	489,466

176,929 individuals (83,235 females) of 489,466 observations; 29,917 alleged offenders, n=45,456

Youth Crime categories: most disadvantaged

	<u>Violent Crime</u>			<u>Property Crime</u>			<u>Drug Crime</u>		
	(1)	(2)	(3)	(4)	(5)	(6)	(4)	(5)	(6)
	All	Male	Female	All	Male	Female	All	Male	Female
Stage 1 DV: Years of Education									
Preparatory Year of Education Treatment	0.834**	0.823**	0.848**	0.834**	0.823**	0.848**	0.834**	0.823**	0.848**
	(0.019)	(0.027)	(0.028)	(0.019)	(0.027)	(0.028)	(0.019)	(0.027)	(0.028)
F-stat. ³	1921.90	899.47	945.62	1921.90	899.47	945.62	1921.90	899.47	945.62
Stage 2 DV: Probability of Offending									
Years of Education	-0.0196*	-0.0210	-0.0058	-0.0273*	-0.0231	-0.0156	-0.0053**	-0.0076**	-0.0023**
	(0.0090)	(0.0147)	(0.0090)	(0.0134)	(0.0211)	(0.0147)	(0.0005)	(0.0008)	(0.0005)
Fixed Effects									
Relative age	yes	yes	yes	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes	yes	yes	yes
Number of Individuals	4322	2276	2046	4322	2276	2046	4322	2276	2046
N	9673	5024	4649	9673	5024	4649	9673	5024	4649
Under identification test ²	936.968	478.275	456.073	936.968	478.275	456.073	936.968	478.275	456.073
Weak identification test ¹	1921.904	899.468	945.622	1921.904	899.468	945.622	1921.904	899.468	945.622
Marginal effects; Standard errors in parentheses; + p<0.10, * p<0.05, ** p<0.01									
¹ Kleibergen-Paap rk LM statistic; ² Kleibergen-Paap rk Wald F statistic									
³ Exceeds F-stat requirements per Lee et. al., 2021									

Youth Crime categories

	<u>Violent Crime</u>				<u>Property Crime</u>				<u>Drug Crime</u>		
	(1)	(2)	(3)		(4)	(5)	(6)		(4)	(5)	(6)
	All	Male	Female		All	Male	Female		All	Male	Female
Stage 1 DV: Years of Education											
Preparatory Year of Education Treatment	0.841**	0.838**	0.843**		0.841**	0.838**	0.843**		0.841**	0.838**	0.843**
	(0.002)	(0.004)	(0.004)		(0.002)	(0.004)	(0.004)		(0.002)	(0.004)	(0.004)
F-stat. ³	1.2e+05	59156.29	57648.72		1.2e+05	59156.29	57648.72		1.2e+05	59156.29	57648.72
Stage 2 DV: Probability of Offending											
Years of Education	-0.0055**	-0.009**	-0.0005		-0.0140**	-0.0187**	-0.0064**		-0.0339**	-0.0464**	-0.0170**
	(0.0007)	(0.0011)	(0.0008)		(0.0011)	(0.0018)	(0.0013)		(0.0010)	(0.0016)	(0.0010)
Fixed Effects											
Relative age	yes	yes	yes		yes	yes	yes		yes	yes	yes
Year	yes	yes	yes		yes	yes	yes		yes	yes	yes
Number of Individuals	176,930	93695	83,235		176,930	93695	83,235		176,930	93695	83,235
N	489466	255373	234,093		489466	255373	234,093		489466	255373	234,093
Under identification test ²	1.2e+04	2.5e+04	2.2e+04		1.2e+04	2.5e+04	2.2e+04		1.2e+04	2.5e+04	2.2e+04
Weak identification test ¹	4.7e+05	5.90e+04	5.8e+04		4.7e+05	5.90e+04	5.8e+04		4.7e+05	5.90e+04	5.8e+04
Marginal effects; Standard errors in parentheses; + p<0.10, * p<0.05, ** p<0.01											
¹ Kleibergen-Paap rk LM statistic; ² Kleibergen-Paap rk Wald F statistic											
³ Exceeds F-stat requirements per Lee et. al., 2021											

End of School:

-0.001** -0.0002

-0.0032** -0.001**

-0.004** -0.0005**

(Beaton et al., 2018)

Youth Crime: persistent offenders

	<u>P(Committing an Offence)</u>			<u>Number of Offences</u>		
	(4)	(5)	(6)	(7)	(8)	(9)
-	All	Male	Female	All	Male	Female
Stage 1 DV: Years of Education						
Preparatory Year of Education Treatment	0.844**	0.835**	0.852**	0.844**	0.835**	0.852**
	(0.002)	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)
F-stat. ³	1.2e+05	58557.85	61531.29	1.2e+05	58557.85	61531.29
Stage 2						
Years of Education	-0.0370**	-0.0533**	-0.0171**	-0.0544**	-0.1557**	-0.0137+
	(0.0011)	(0.0018)	(0.0011)	(0.0102)	(0.0171)	(0.0081)
Fixed Effects						
Year	yes	yes	yes	yes	yes	yes
Relative age	yes	yes	yes	yes	yes	yes
Number of Individuals	162886	84480	78406	162886	84480	78406
N	465236	238657	226579	465236	238657	226579
Under identification test ¹	1.2e+05	2.3e+04	2.2e+04	1.2e+05	2.3e+04	2.2e+04
Weak identification test ²	4.5e+04	5.90e+04	6.2e+04	4.5e+04	5.90e+04	6.2e+04
Marginal effects; Standard errors in parentheses; + p<0.10, * p<0.05, ** p<0.01						
¹ Kleibergen-Paap rk LM statistic; ² Kleibergen-Paap rk Wald F statistic						
³ Exceeds F-stat requirements per Lee et. al., 2021						

Youth Crime categories: persistent offenders

	<u>Violent Crime</u>			<u>Property Crime</u>			<u>Drug Crime</u>		
	(1)	(2)	(3)	(4)	(5)	(6)	(4)	(5)	(6)
	All	Male	Female	All	Male	Female	All	Male	Female
Stage 1 DV: Years of Education									
Preparatory Year of Education Treatment	0.844**	0.835**	0.852**	0.844**	0.835**	0.852**	0.844**	0.835**	0.852**
	(0.002)	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)	(0.002)	(0.004)	(0.003)
F-stat. ³	1.2e+05	58557.85	61531.29	1.2e+05	58557.85	61531.29	1.2e+05	58557.85	61531.29
Stage 2 DV: Probability of Offending									
Years of Education	-0.0017**	-0.0026**	-0.0004	-0.0051**	-0.0078**	-0.0014*	-0.0053**	-0.0076**	-0.0023**
	(0.0004)	(0.0007)	(0.0005)	(0.0007)	(0.0012)	(0.0007)	(0.0005)	(0.0008)	(0.0005)
Fixed Effects									
Relative age	yes	yes	yes	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes	yes	yes	yes
Number of Individuals	162886	84480	78406	162886	84480	78406	162886	84480	78406
N	465236	238657	226579	465236	238657	226579	465236	238657	226579
Under identification test ²	1.2e+05	2.3e+04	2.2e+04	1.2e+05	2.3e+04	2.2e+04	1.2e+05	2.3e+04	2.2e+04
Weak identification test ¹	4.5e+04	5.90e+04	6.2e+04	4.5e+04	5.90e+04	6.2e+04	4.5e+04	5.90e+04	6.2e+04
Marginal effects; Standard errors in parentheses; + p<0.10, * p<0.05, ** p<0.01									
¹ Kleibergen-Paap rk LM statistic; ² Kleibergen-Paap rk Wald F statistic									
³ Exceeds F-stat requirements per Lee et. al., 2021									

End of School:

-0.000 -0.0007

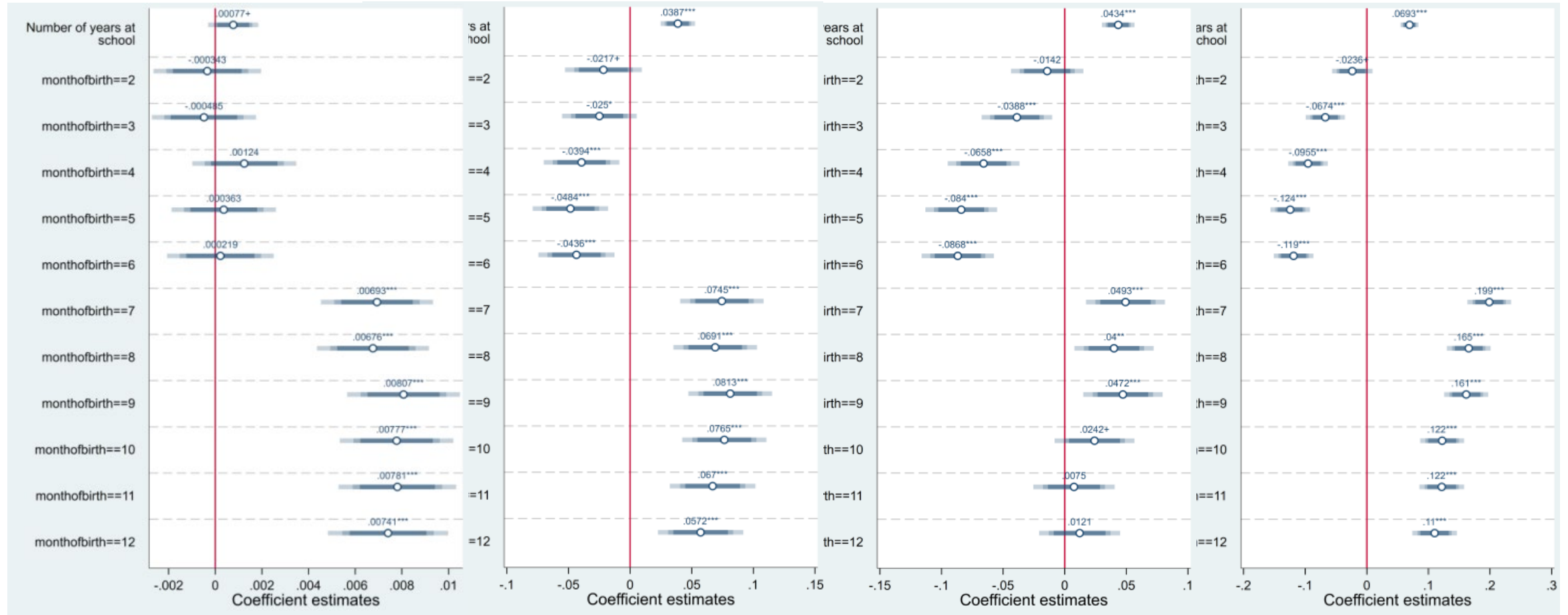
-0.0005 -0.002

-0.0003 -0.001

(Beaton et al., 2018)

Prep: extra year at the beginning of schooling

Relative Age Effects: English Year 3



Attendance

Behaviour

Effort

Achievement

Relative age

The Economics of Crime Approach

SUCCESS PROBABILITY:
GAINS FROM ILLEGAL WORK

DETECTION:
SANCTIONS

GAINS FROM LEGAL
WORK OR ACTIVITY

$$(1 - \pi)U(W_c) - \pi U(S) > U(W_L)$$

Economic returns to crime

- Criminal earnings
- Value of loot
- Security responses

Deterrence and the criminal justice system

- Policing/enforcement
- Sentencing/sanctions

Legal alternatives to crime

- Labour market
- Education/skills
- Careers

Becker/Ehrlich (1960) Utility Model

(Machin, 2014)

Parent Response to > Suspensions & > Exclusions

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SYSTEMDOWN

SCHOOLS PUT UP EXCLUSION ZONE

SIX MONTHS AFTER REVIEW RECOMMENDED URGENT CHANGES, PARENTS OF CHILDREN WITH DISABILITY STILL STRUGGLE TO KEEP KIDS IN CLASSES

EXCLUSIVE
EMMALINE STIGWOOD

QUEENSLAND students with a disability are being suspended and excluded from school in huge numbers despite a landmark review recommending urgent changes.

Six months after an investigation revealed poor outcomes and high suspension rates, parents have spoken of the on-going reality of life in the classrooms. They claim the system is still failing their children, with schools refusing entry and medical opinions being ignored.

One mother, whose 13-year-old son with autism spectrum disorder has lost 210 school days to suspension, was told she was lucky he hadn't been excluded for longer.

Another mother is taking legal action after her son, diagnosed with ASD and other disabilities, bounced between different schools where he was allegedly repeatedly attacked and made to sit in an "exit chair" in class.

Such strategies to deal with behaviour were called out by the exhaustive Review of Education for Students with Disability in Queensland State Schools that in February recommended immediate changes.

The report found that short suspensions, up to 10 days, for students with a disability were

growing and hit almost 7500 in 2015. About another 1500 longer suspensions and exclusions were given out.

The latest 2016 data also reveals the number of suspensions for prep children had doubled in three years but no breakdown was given on how many of them had a disability.

The Department of Education and Training could not provide any figures for how many students with a disability had been suspended during the past 18 months, despite it now requiring school principals to provide measures and be judged on how they reduced their suspension figures.

Sharon Horan of Autism

Behavioural Intervention Queensland said the number of children pushed into home schooling or distance education and those repeatedly suspended because of behaviour linked to disability was unacceptably high.

She said some schools were turning away students with ASD or making it so difficult to enrol that parents gave up. When parents did secure a spot they didn't feel comfortable making a complaint about a problem, yet that was often the only way to get expert help from the government-funded Autism Hub and Reading Centre.

"Even if they do complain and it's followed up there is no

guarantee the recommendations made by Autism Hub will be followed," she said.

The Department of Education and Training and ASD advocates said it was going to take time for changes to occur in a system that dealt with 80 per cent of all students with a disability in Queensland.

As well as all state school principals now being measured on reducing suspension rates, it has set three other targets on lifting A-E grades of students, increasing the proportion receiving high school certificates and cutting the number not attending fulltime. Heads of special education will be elevated to deputy principal level.

Year	Total Suspensions	Short Suspension	Long Suspension
2011	~6500	~5500	~1000
2012	~7000	~6000	~1000
2013	~7500	~6500	~1000
2014	~8000	~7000	~1000
2015	~8500	~7500	~1000

SOURCE: QUEENSLAND EDUCATION

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Sunday Mail | SPECIAL INVESTIGATION

Suspensions rob students of chance for education

EMMALINE STIGWOOD

CLASS IS OUT: Annette Denham and Qwyn, 13, who has lost 210 days of school; and (bottom right) Andrea Ford with Josh, 13, who is now doing distance education. Main picture: AAP/Sarah Marshall

ANNETTE Denham's son has already lost 210 days of his education to suspension, and now she fears he will never be able to catch up.

But she refuses to give up on him being included as part of the state school system.

Qwyn Denham, 13, has ASD and his behaviour can be challenging and disruptive but his mother said repeated suspensions without making changes at school or classroom level were pointless.

Within three weeks of starting at River Lake State High School this year, Qwyn was suspended for kicking a boy during a basketball game.

In the third term, he was suspended after a basketball game incident escalated. He was back at school for only two days when it was suggested he stay home again.

"Because he has missed so much school he's in the situation where he is floundering but instead of making him feel like a failure, why don't we look at ways to solve the

underlying issue," Ms Denham said. "I'm not giving up because this is a kid who, if things continue, in five years will leave school without the skills needed."

The issues are the same for North Lakes mother Andrea Ford, whose son Josh, 13, was also diagnosed with ASD and other disabilities.

She has lodged a complaint with the Human Rights Commission over his treatment in state schools after a two-year battle to find him a safe place at schools at Bracken Ridge, Mango Hill and Pine Rivers.

She is also considering filing a personal injury claim related to Josh being beaten in the playground and being made to sit in an "exit chair".

Within three weeks of starting high school this year she had pulled him from class due to injuries and escalating threats to hurt himself. He's now doing distance education at the family's expense.

"My son is not a gap in the system, he is a child with a right to an education," she said.

Assistant director-general of the newly formed Disability and Inclusion branch, Deb Dunstone said a drop in suspensions and improved results would be seen year on year.

Education Minister Kate Jones said parents of children with a disability had every

right to expect full and tailored education.

"We knew this review would find things and we would have to lift our game but the reforms are not going to be a fairy daisy. It's going to be tangible reforms and we are going to see a result," she said.

Some schools were turning away students with ASD or making it so difficult to enrol that parents gave up

